

Customer behavior prediction platform

A trained customer analytics prototype with a Streamlit dashboard, FastAPI backend and reproducible Python 3.11 pipeline.

The churn model meets the required AUC and recall targets. The validation-selected customer-value model misses its R^2 target, and the LSTM does not beat its static baseline. These limitations remain visible in the application.

Measure	Held-out result	Acceptance
Churn ROC-AUC	0.8168	0.80 target met
Churn recall	89.60%	0.70 target met
90-day value R^2	0.4820	0.60 target not met
Source coverage	93.7%	70% target
Evaluation cohort	788 test customers	Disjoint customer IDs

Delivered scope

The package includes both workbook sheets, clean data, feature and sequence tables, six classical churn models, ANN/LSTM/autoencoder artifacts, value and category models, two clustering approaches, SHAP/LIME explanations, dashboard, API, tests, notebooks, diagrams and Docker configuration.

Runtime boundary

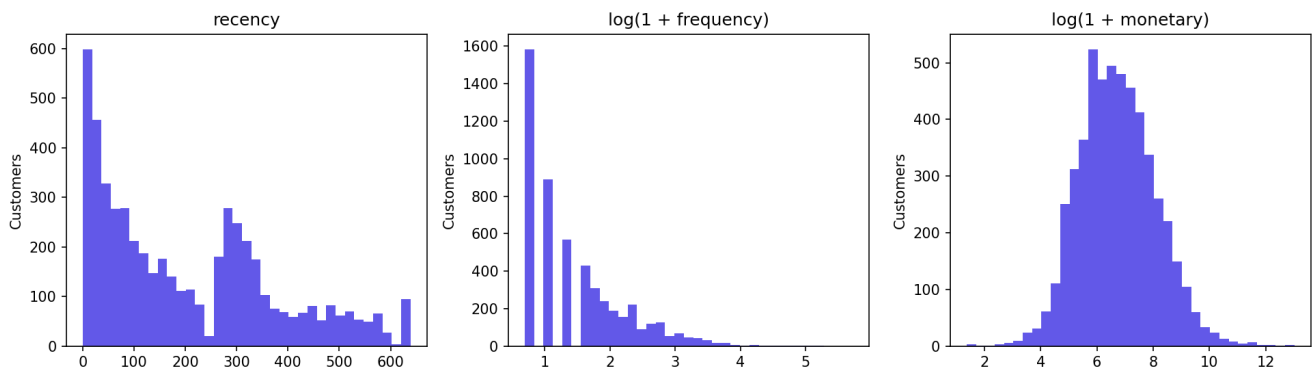
Training and application execution use CPython 3.11.13 on a Linux CPU. Native API persistence is tested through SQLite. PostgreSQL schema and Compose configuration are included, but Docker is unavailable and local PostgreSQL initialization was blocked. Clean-machine container acceptance remains open. Windows launch instructions are provided but were not executed on a Windows host.

This is a historical prediction snapshot at 1 September 2011. A 90-day gross-spend estimate is not lifetime profit. UI scores include training/validation customers; only explicitly held-out tables measure generalization.

Data foundation

Audit item	Observed
Source transaction lines	1,067,371
Exact duplicate rows	34,335
Invalid price / zero quantity	6,019
Extreme lines excluded	7
Clean ledger	1,027,010
Missing Customer ID	22.77%
Eligible snapshot customers	5249

Returns retain signed negative amounts and produce a return-rate feature. IQR flags do not automatically remove wholesale purchases. Domain bounds remove absolute quantities above 50,000 and prices above GBP 20,000; original bytes and quarantined records are preserved. Exact identical invoice lines cannot be distinguished from legitimate repeated lines, which is a source limitation.



Anonymous records remain in the clean ledger but cannot join customer models. Administrative StockCodes are excluded from product features while retained in accounting. Descriptions are normalized with a per-SKU modal lookup fitted only before cutoff. Correlation, country, seasonality and VIF assets accompany the notebooks.

Actual source dates run from December 2009 to December 2011 and do not span a leap year. Churn prevalence is 57.31%, so the PRD minority-class assumption is not true for this snapshot.

Evaluation design and churn

Features use timestamps strictly earlier than 2011-09-01. Labels cover [2011-09-01, 2011-11-30), a complete 90-day interval. The 70/15/15 stratified split has 3,674 training, 787 validation and 788 test customers, seed 42. No customer overlaps partitions. All learned encoders/scalers are refit within training CV folds.

Model	CV AUC	Test AUC	Recall	Precision	F1	Accuracy
XGBoost	0.801	0.818	0.918	0.691	0.788	0.717
RandomForest	0.800	0.817	0.896	0.711	0.793	0.731
ANN	0.800	0.817	0.852	0.725	0.783	0.730
LogisticRegression	0.799	0.814	0.905	0.706	0.793	0.730
LightGBM	0.795	0.813	0.878	0.718	0.790	0.732
KNN	0.792	0.805	0.894	0.710	0.791	0.730
DecisionTree	0.784	0.794	0.940	0.667	0.781	0.697

Random Forest wins validation AUC (0.7805). Its 36% threshold maximizes validation F1 subject to recall ≥ 0.70 . Test confusion matrix: TN=171, FP=165, FN=47, TP=405. A slightly higher XGBoost test AUC does not justify retrospective model switching.

Random Forest and XGBoost searches use five-fold training CV. XGBoost early stopping uses validation. Neural CV reserves an internal stopping split separate from the scored fold. ANN CV uses up to 15 epochs and the final fit up to 35; validation loss controls early stopping.

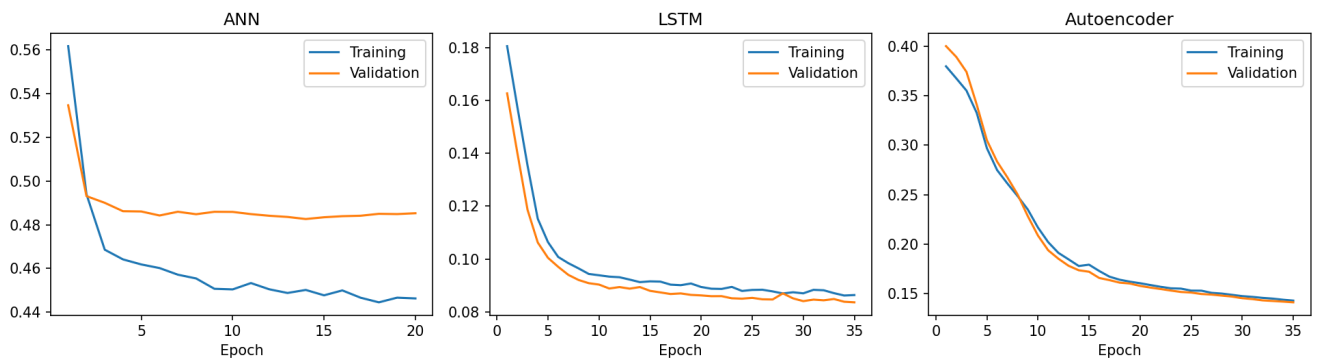
The customer-holdout design follows the PRD but does not validate future-time drift. Shared pre-cutoff catalog statistics are operationally known. A genuinely later untouched period is needed for temporal deployment claims.

Customer value and purchase behavior

Value model	Val R ²	Test R ²	MAE GBP	RMSE GBP
Ridge	0.273	0.655	607.48	3,611.96
RandomForestRegressor	0.848	0.482	709.99	4,425.68
LightGBMRegressor	0.669	0.284	777.32	5,202.69

The regression target is untrimmed future 90-day gross spend. Random Forest wins validation but generalizes less well on rare wholesale outcomes. Ridge exceeds 0.60 test R² but was not selected using validation, so it is not promoted after test inspection. No high-error customers are removed to improve the reported score.

LSTM capped-wait test MAE is 20.56 days, compared with 19.40 for Ridge. The neural sequence model has not demonstrated improvement over static features.

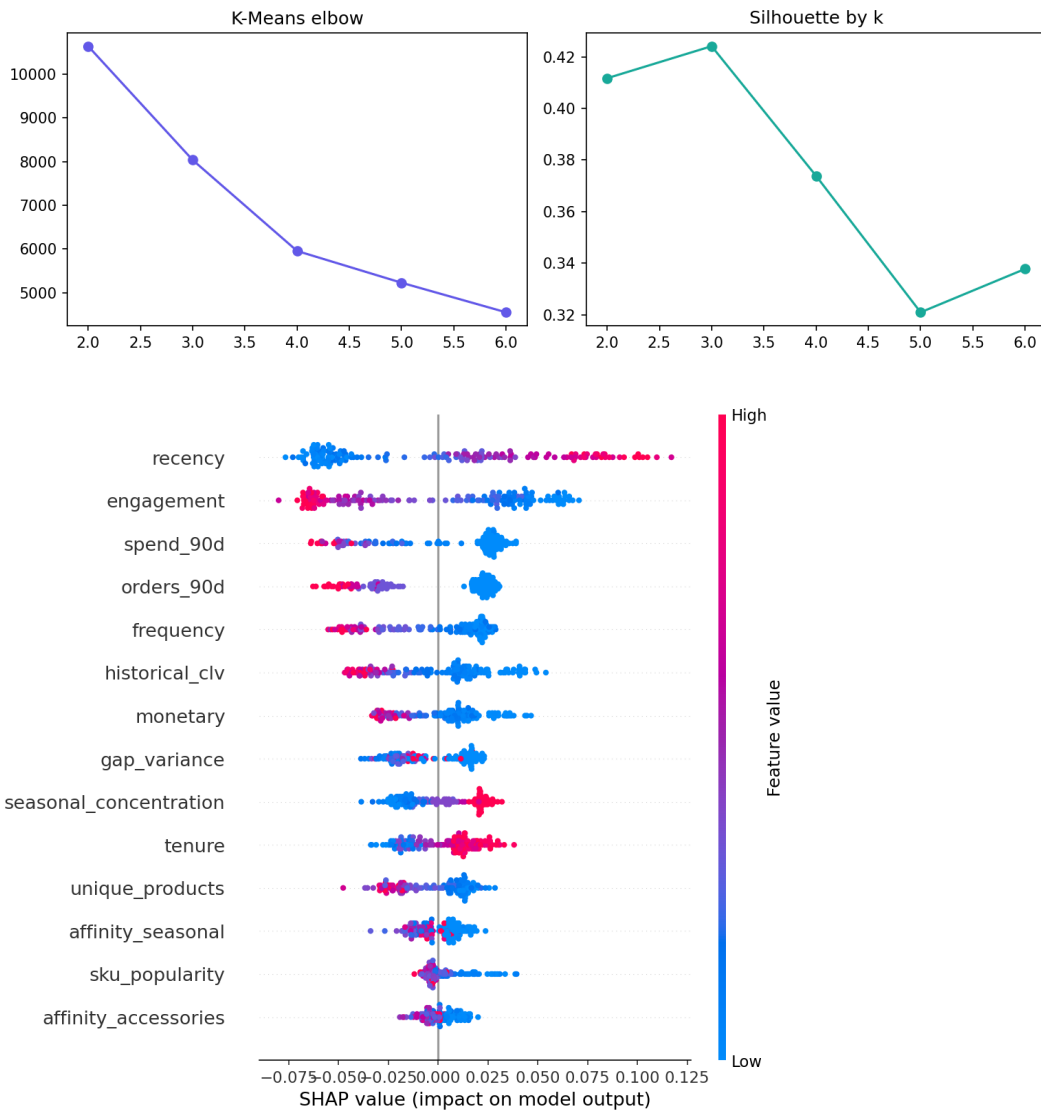


Next proxy-category classification has test accuracy 0.551 and macro-F1 0.325. Purchase probability equals one minus churn probability for the same outcome horizon. Recommendations suggest unseen popular products in the predicted description-derived category. Ranking quality is not evaluated.

Segmentation and explainability

K-Means chooses $k=3$ from 2-6 using training silhouette, supported by an elbow plot.

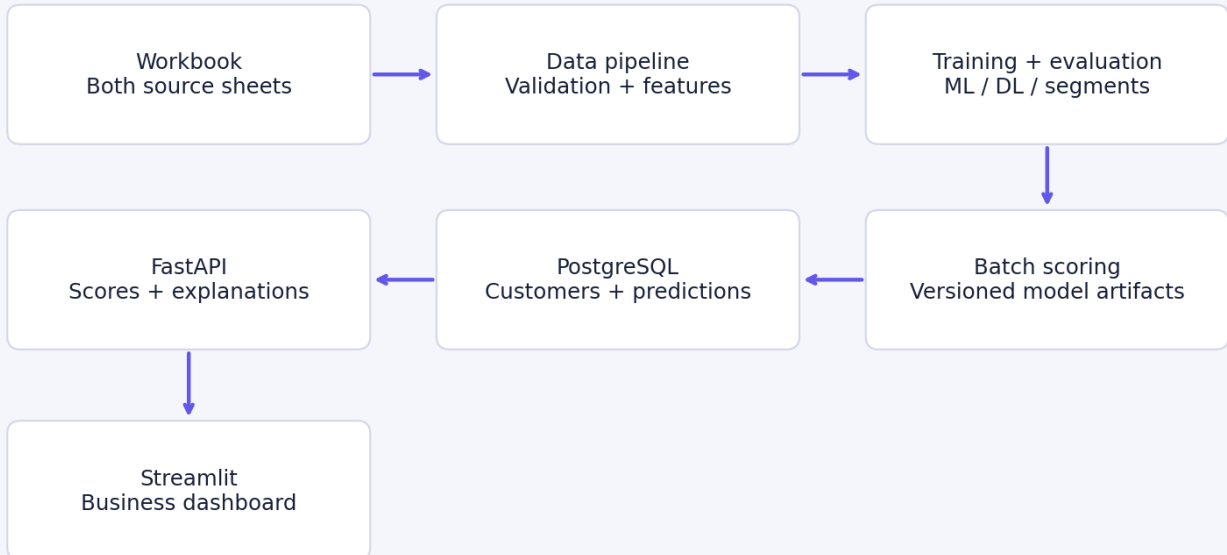
Silhouette=0.424; Davies-Bouldin=0.802. The Gaussian mixture comparison has silhouette=0.163 and Davies-Bouldin=2.100. Profiles use historical activity/value; individual members may differ from cluster medians.



The API supplies SHAP and LIME for any saved customer. Three low/borderline/high-risk examples include force HTML files, waterfall PNGs and JSON explanations. Positive contributions raise predicted risk relative to the model baseline. LIME is an approximation with a reported local fit score. Attribution is not a causal effect of offering a discount.

Architecture and application

Vantara customer intelligence architecture



Streamlit is an HTTP-only client: it never calls a model directly. The API loads immutable artifacts, reads stored customer scores and produces on-demand explanations. Full feature CSVs run new tabular inference and persist batch history; ID-only CSVs retrieve the saved snapshot.

The relational schema contains customers, predictions, segments and invoice-level transactions, plus `batch_scores`. The clean line-item ledger remains in Parquet. First startup seeds the historical snapshot idempotently; intentional retraining requires an explicit database refresh. Scoring and query behavior is covered through the same SQLAlchemy schema with SQLite.

The dashboard offers an overview, segment/country/value filters, high-risk value priorities, customer search and explanations, model metrics, CSV uploads/downloads and customer PDFs. A three-month moving-average revenue overlay is labeled as a baseline. It has no validated forecast interval.

Testing and acceptance

Check	Evidence
Source test coverage	93.7%
Reproducibility	Full retraining in an isolated temporary directory
API contracts	Health, validation, lookup, batch persistence and downloads
Live UI	Five HTTP-backed Streamlit page tests and Chromium upload/download journey
Saved lookup p95	2.50 ms, 100 local requests
Initial dashboard load	3.59 seconds
Warm segment navigation	0.70 seconds
Docker and PostgreSQL	Configuration implemented; clean-machine execution unverified

Tests exercise signed returns, RFM arithmetic, future-data perturbations, exact date boundaries, padded sequences, disjoint IDs, malformed CSV, unseen countries, explanations, serialized inference and database persistence. Full retraining uses identical frozen settings and checks saved probabilities/value/timing predictions; it never tunes using the test partition.

Local performance is not a production SLA. Measurements use warm SQLite and a CPU browser session, not concurrent PostgreSQL traffic. Initial loading can exceed the 3-second target; warm segment navigation is reported separately. Explanation latency is outside the stored-score lookup benchmark.

Remaining work

The selected CLV model misses its target and the LSTM misses baseline improvement. No labeled fraud performance exists. Future work needs temporal validation, value-model robustness, calibrated probabilities and real category/discount data. Authentication, role-based access, CI/CD, streaming, A/B tests and Transformers remain deferred as allowed by the PRD.

Mathematical appendix 1

Features and classical models

Customer summaries

For customer i at cutoff c , the history contains only timestamps $t < c$. Recency $R=(c-\max t)/\text{day}$; frequency F counts distinct positive invoices; gross monetary M sums positive quantity times price. Historical net value includes signed returns. Recent features restrict the same history to 90 days.

Trend uses six chronological 30-day counts n_j . Centered time $x_j=j-2.5$ gives the least-squares slope $\text{sum}(x_j n_j)/17.5$. Gap variance is the sample variance of consecutive inter-order days. Engagement $= 100/3 [\exp(-R/90)+1-\exp(-F/10)+1-\exp(-M/2500)]$. This composite is fixed, not fitted using future outcomes.

Logistic regression and Ridge

The churn logit $z=w^T x+b$ produces $p=1/(1+\exp(-z))$. Weighted cross-entropy is $-\text{mean}[y \log(p)+b(1-y)\log(1-p)]$, plus an L2 regularization penalty. Balanced weights come from training prevalence. C controls inverse regularization. Ridge minimizes squared regression error plus α times the sum of squared coefficients.

Correlated RFM components and category shares can have high or infinite VIF. The pipeline exports VIF and retains these PRD feature families for tree models and regularized baselines. Their coefficients should not be interpreted as causal effects.

Preprocessing and ensembles

StandardScaler computes $z_j=(x_j-\mu_j)/\sigma_j$ with statistics fit only on training folds. Median imputation and one-hot country encoding are persisted. Unknown countries are ignored. Random Forest averages randomized trees, while boosting adds weak learners along objective gradients. XGBoost stopping uses validation rather than test records.

Mathematical appendix 2

Neural and sequential models

ANN optimization

The ANN combines affine layers, ReLU, batch normalization and dropout, ending in a churn logit. Backpropagation applies the chain rule to compute gradients through each layer. Adam updates weights using running first and second gradient moments; the delivered learning rate is 0.001 and weight decay 0.0001.

BCEWithLogits combines sigmoid and cross-entropy stably. Positive-class weight is negative-count/positive-count in the training rows. Dropout is active only during training. Early stopping restores the state with minimum validation loss after six non-improving epochs. Curves for training/validation are included for all neural models.

Purchase-event LSTM

Each sequence has at most 12 product invoices. Event inputs contain log amount, log gap, age at cutoff and five proxy-category shares. Packed lengths ensure zero padding does not become a fabricated event. The LSTM maintains hidden h_t and memory c_t , with sigmoid input/forget/output gates and a tanh candidate memory update.

The final hidden state enters dropout and a sigmoid head. Output times 90 predicts $\min(\text{waiting time}, 90)$. Mean squared error trains the normalized target. Customers without a future purchase receive the 90-day cap, so this is not uncensored survival estimation.

Spending autoencoder

The network compresses d inputs through $d \rightarrow 16 \rightarrow 5$, then reconstructs through $5 \rightarrow 16 \rightarrow d$. Per-customer reconstruction error is $\text{mean}_j[(x_j - \text{reconstructed}_x_j)^2]$. A validation 95th-percentile threshold requests manual review. Without fraud labels, an anomaly flag cannot establish fraud precision or recall.

Mathematical appendix 3

Evaluation and interpretation

Metrics

Precision= $TP/(TP+FP)$, recall= $TP/(TP+FN)$, $F1=2 \text{ precision recall}/(\text{precision}+\text{recall})$, and accuracy= $(TP+TN)/N$. ROC-AUC measures ranking over thresholds. Confusion-matrix rows are actual purchase/churn and columns are predicted purchase/churn. Validation fixes the deployed decision threshold before test scoring.

$MAE=\text{mean}(\text{abs}(y-\text{prediction}))$; $RMSE=\sqrt{\text{mean}((y-\text{prediction})^2)}$;

$R^2=1-\text{sum}((y-\text{prediction})^2)/\text{sum}((y-\text{mean}(y))^2)$. R^2 can be negative. Large wholesale purchases dominate squared error, so MAE/RMSE remain reported in original currency without target trimming.

Clustering and attributions

K-Means minimizes within-cluster squared Euclidean distances on standardized log features.

Silhouette= $(b-a)/\max(a,b)$, where a is within-cluster distance and b the nearest alternative-cluster distance; higher is better. Davies-Bouldin compares within-cluster scatter to separation; lower is better. Gaussian mixtures allow overlapping elliptical groups.

SHAP distributes model-output differences relative to a baseline across input features. TreeSHAP for the deployed Random Forest explains churn probability. LIME fits an interpretable local surrogate and exposes its local R^2 . Neither predicts the causal effect of a retention campaign.

Provenance

Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. DOI 10.24432/C5CG6D. <https://archive.ics.uci.edu/dataset/502/online+retail+ii>. CC BY 4.0. Source bytes are the supplied workbook. Requirements are the supplied Vantara PRD v1.0. Metadata records versions, configuration, seed and the feature-table SHA-256 hash.

Implementation guidance: scikit-learn Common pitfalls and recommended practices, https://scikit-learn.org/stable/common_pitfalls.html.